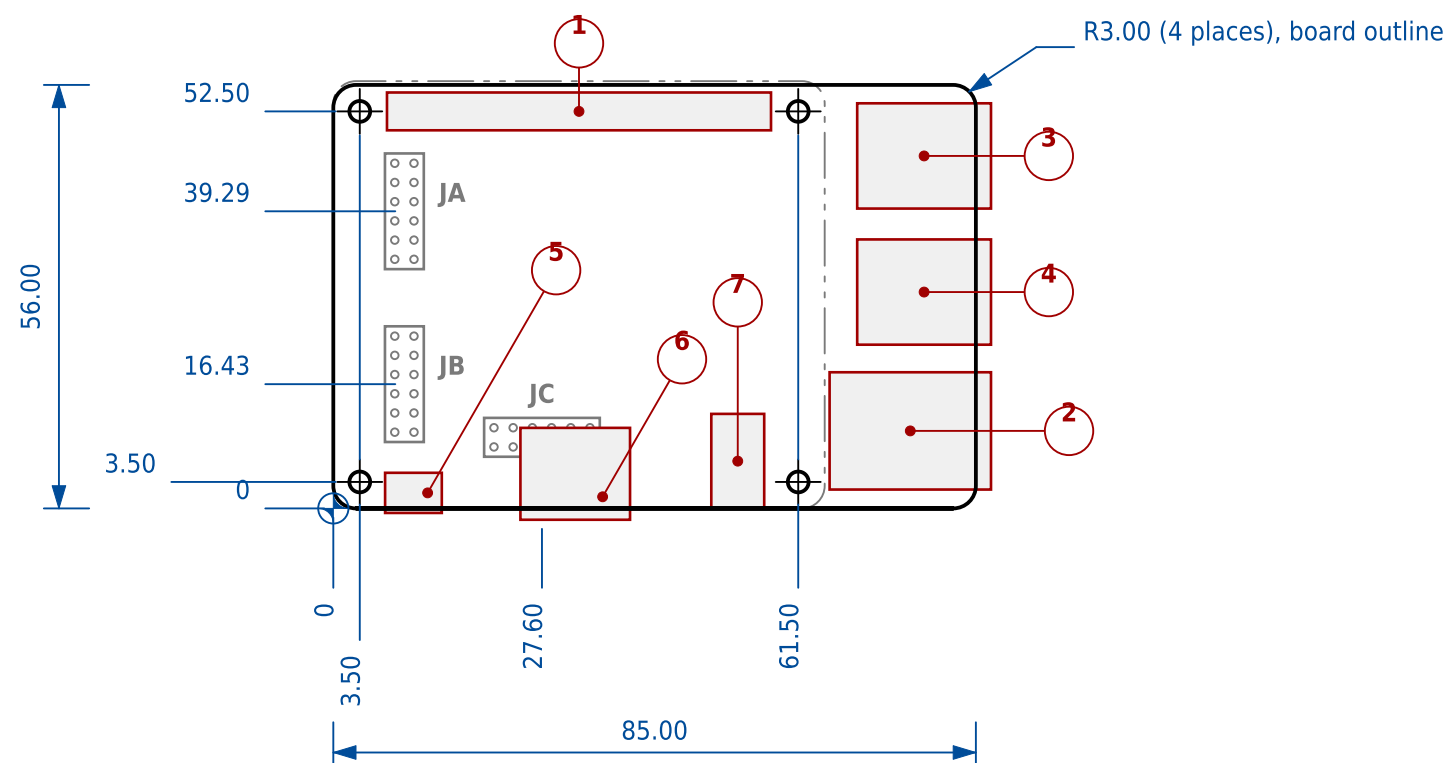




## NOTES

1. All dimensions in millimetres. The datum symbol marks the origin: the lower-left corner of the board outline, X right, Y up, viewed from the component side.
2. Phantom outline is the Digilent Pmod HAT Adapter fitted to the 40-pin GPIO header, drawn in this board's frame because its mounting holes coincide with this board's.
3. Connector outlines are the component body as drawn by Raspberry Pi Ltd, overhang past the board edge included.
4. Mounting hole diameter: Carried over from the Pi 3 Model B drawing, which states 4x M2.5 holes drilled to 2.75 +/-0.05 mm; the two boards share an identical outline, hole pattern and DXF geometry. The 3B+ drawing gives neither a hole size nor a tolerance, so the +/-0.05 is not repeated.
5. Hole IDs are assigned by this drawing, bottom row first then left to right, and mean the same thing on every Raspberry Pi sheet here.
6. Design mounting hole keep-outs to 6.2 mm diameter on every model. That is what the Raspberry Pi HAT mechanical specification ([github.com/raspberrypi/hats](https://github.com/raspberrypi/hats)) requires around each hole, and it is the largest of the figures the individual models publish. The KEEPOUT column gives what this model's own drawing shows, which is not always anything.
7. Hole positions are the 3.5 mm inset and 58 x 49 mm rectangle dimensioned on this model's drawing.

## NOTES (continued)

8. Pmod host JC on the Pmod HAT Adapter overhangs the lower board edge in the same region as this board's HDMI and power connectors. Digilent's own reference manual warns to fit the two standoffs opposite the 40-pin connector so that JC's pins cannot touch the HDMI shell.
9. Those host positions are DERIVED, not published: Digilent issue no mechanical drawing for the adapter. Good to about  $\pm 0.75$  mm. JA and JB face out of the left edge, JC out of the lower edge, all right-angle hosts whose bodies overhang the edge. The Pmod HAT Adapter sheet has the derivation.
10. Pmod host coordinates are tabulated to three decimals, unlike everything else on this sheet. That is not a claim to micrometre accuracy: rounded to two, two adjacent hosts print 22.85 apart and contradict the 22.86 pitch the specification mandates and this drawing dimensions.
11. Pmod host ports are lettered JA, JB, JC on the Pmod HAT Adapter and on the Raspberry Pi sheets, after Digilent's own labelling; named by signal direction on the Tiny Tapeout demo boards, after the silkscreen; and numbered PMOD 1 to 3 on the mounting plate sheets, which number positions rather than ports. The fitting guide TT-MP-02 maps a board's first host to a plate position.

## HOLE SCHEDULE

| ID  | X mm  | Y mm  | DIA mm | KEEPOUT mm |
|-----|-------|-------|--------|------------|
| MT1 | 3.50  | 3.50  | 2.75   | not given  |
| MT2 | 61.50 | 3.50  | 2.75   | not given  |
| MT3 | 3.50  | 52.50 | 2.75   | not given  |
| MT4 | 61.50 | 52.50 | 2.75   | not given  |

## FEATURE SCHEDULE

| # | FEATURE                     | X EXTENT mm    | Y EXTENT mm    |
|---|-----------------------------|----------------|----------------|
| 1 | 40-pin GPIO header          | 7.10 to 57.90  | 50.00 to 55.00 |
| 2 | Ethernet RJ45               | 65.65 to 87.00 | 2.50 to 18.00  |
| 3 | USB 2.0 type A (upper pair) | 69.30 to 87.00 | 39.65 to 53.50 |
| 4 | USB 2.0 type A (lower pair) | 69.30 to 87.00 | 21.65 to 35.50 |
| 5 | micro-USB power input       | 6.85 to 14.35  | -0.60 to 4.75  |
| 6 | HDMI type A                 | 24.75 to 39.25 | -1.50 to 10.65 |
| 7 | 3.5 mm A/V jack             | 50.00 to 57.00 | 0.00 to 12.50  |

## PMOD HAT ADAPTER HOSTS

| PORT | EDGE        | CX mm  | CY mm  | PIN1 X mm | PIN1 Y mm |
|------|-------------|--------|--------|-----------|-----------|
| JA   | left edge   | 9.400  | 39.290 | 8.130     | 45.640    |
| JB   | left edge   | 9.400  | 16.430 | 8.130     | 22.780    |
| JC   | bottom edge | 27.600 | 9.400  | 21.250    | 8.130     |

## SOURCES

1. Mechanical drawing:  
<https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-plus-mechanical-drawing.dxf> - Raspberry Pi Ltd
2. Mounting holes:  
<https://datasheets.raspberrypi.com/rpi3/raspberry-pi-3-b-plus-mechanical-drawing.pdf> - Carried over from the Pi 3 Model B drawing, which states 4x M2.5 holes drilled to 2.75 +/-0.05 mm; the two boards share an identical outline, hole pattern and DXF geometry. The 3B+ drawing gives neither a hole size nor a tolerance, so the +/-0.05 is not repeated.

## RASPBERRY PI 3 MODEL B+

85 x 56 mm

|                                              |                           |                                                                            |                     |
|----------------------------------------------|---------------------------|----------------------------------------------------------------------------|---------------------|
| DRAWN<br><b>generated</b>                    | DATE<br><b>2026-09-10</b> | UNITS<br><b>mm</b>                                                         | SCALE<br><b>1:1</b> |
| SIZE<br><b>A3</b>                            | SHEET<br><b>1 OF 1</b>    | DRAWING NO<br><b>RPI-03</b>                                                | REV<br><b>A</b>     |
| MATERIAL<br><b>PCB, thickness not stated</b> |                           | GENERAL TOLERANCE<br><b>edge +/-0.20 hole pos +/-0.10 hole dia +/-0.08</b> |                     |